FOUNDATION PROGRAMME / AGES 9–12

Young Problem Solvers

My Coursebook — Volume 1

Unit 1 · Think with Numbers | Weeks 1–3 | Six 90-minute sessions

Pilot Version 1.0 · Intended for educator review and testing; not an accredited syllabus or validated intervention.

Welcome to Think with Numbers

This coursebook explains the important ideas for six lessons. Bring it to class, read the examples, and use drawings or number sentences to explain your thinking. The separate practice workbook gives you space to work independently.

Our problem-solving routine

- DISCOVER — What is being asked?
- LEARN — Which new idea or method will help?
- SOLVE — Show the mathematics, not only an answer.
- EXPLAIN — Say how you know and compare methods.
- REFLECT — Check and improve your work.

Our three-week journey

Week	Big question	New abilities
1	How do we know an answer is sensible?	Place value; order; rounding; exact and estimated addition
2	How can we choose within a budget?	Reserves; quantities; optional extras; trade-offs
3	How do groups and prices affect choices?	Multiplication; division; remainders; total packet cost

Using this book

For each session: read the Big Idea, study the worked example, attempt the two guided questions, and discuss the real-life application. Your instructor will give feedback and independent practice from the companion workbook.

WEEK 1 / SESSION 1 / YPS-F1-U1-L01

Number Detectives: Place Value

How does the position of a digit change its value?

I can …

- Represent five-digit amounts using ten-thousands, thousands, hundreds, tens and ones.
- Compare and order amounts without guessing from the first digit alone.
- Explain a comparison using place-value language.

Learn the idea

The value of a digit depends on its position. To compare whole numbers, begin with the highest place value and move right until you find a difference.

Worked example — follow the reasoning

UGX 24,650 has 2 ten-thousands, 4 thousands, 6 hundreds, 5 tens and 0 ones. To compare 24,650 and 24,560: both have 2 ten-thousands and 4 thousands; 6 hundreds is more than 5 hundreds. So 24,650 > 24,560.

- Write each amount in a place-value table.
- Check the ten-thousands column, then thousands, then hundreds.
- Write >, < or =, and explain the first place where the numbers differ.

Try together

Which is larger: UGX 18,950 or UGX 19,050? Explain.

_____

_____

Partition 32,406 as a sum of place values.

_____

_____

Where could we use this?

Pairs act as shopkeepers who must audit incorrectly ordered price tags.

A useful sentence starter

“I chose this method because _____. I checked it by _____.”

Words to remember

place value · estimate · exact · budget · reserve · required · optional · remainder · packet · total cost

WEEK 1 / SESSION 2 / YPS-F1-U1-L02

Estimate, Calculate, Check

How can an estimate help us catch mistakes?

I can …

- Round whole-number prices to the nearest 100 or 1,000.
- Estimate a sum, then calculate it exactly.
- Assess whether an answer is reasonable and check it using another strategy.

Learn the idea

Estimation gives a sensible range before an exact calculation. When rounding to the nearest 1,000, look at the hundreds digit; when rounding to the nearest 100, look at the tens digit.

Worked example — follow the reasoning

UGX 4,800 + UGX 6,300: nearest 1,000 gives 5,000 + 6,000 = 11,000. Exact sum: 11,100. A claimed total of 12,100 is too high by UGX 1,000. Nearest 100 gives 4,800 + 6,300 = 11,100.

- Decide how precise the estimate needs to be.
- Round both amounts using the correct next digit.
- Add exactly and compare with the estimate.
- Use subtraction or regrouping to verify.

Try together

Estimate and calculate 7,480 + 2,620 to the nearest thousand.

A child writes 3,950 + 4,180 = 9,130. How would you check?

Where could we use this?

Teams audit three fictional receipts; each individual records one independent correction.

A useful sentence starter

“I chose this method because _____. I checked it by _____.”

Words to remember

place value · estimate · exact · budget · reserve · required · optional · remainder · packet · total cost

WEEK 2 / SESSION 1 / YPS-F1-U1-L03

Budgets and Priorities

What must we pay for before we choose extras?

I can …

- Distinguish total budget, reserved money, required costs and optional purchases.
- Compute an accurate subtotal using multiplication and addition.
- Decide how much remains available without spending reserved money.

Learn the idea

A budget is a limit. Reserved money is not available for optional purchases. Calculate required purchases before deciding about extras.

Worked example — follow the reasoning

Budget UGX 25,000; reserve UGX 3,000. Supplies limit = 22,000. Required: notebooks 2 × 2,500 = 5,000; pencils 3 × 1,000 = 3,000; tape 4,000; poster 2,000. Required supplies = 14,000. Optional supplies allowance = 22,000 − 14,000 = 8,000.

- Write the full budget and the reserve separately.
- Multiply quantity × price for each required item.
- Add required costs, then subtract from supplies limit.
- Label the remaining amount precisely: available for extras.

Try together

If the reserve rises to UGX 5,000, what is the new supplies limit?

Could we buy an extra poster for UGX 2,000 and one marker for UGX 3,500 after the required supplies?

Where could we use this?

Pairs build a feasible supply list, then each learner must independently prove affordability.

A useful sentence starter

“I chose this method because _____. I checked it by _____.”

Words to remember

budget · required purchase · optional purchase · reserve · constraint · trade-off

WEEK 2 / SESSION 2 / YPS-F1-U1-L04

Choices, Trade-offs and New Information

Which affordable plan best meets a goal?

I can …

- Compare alternative shopping plans against an explicit budget constraint.
- Explain why an affordable plan may still be less suitable than another.
- Revise a plan when a price, reserve or requirement changes.

Learn the idea

A good choice is more than the cheapest price. First reject plans that break constraints; then compare how well affordable options meet the goal and keep enough reserve.

Worked example — follow the reasoning

After required supplies UGX 14,000 and reserve UGX 3,000, optional allowance is UGX 8,000. Plan A: two markers 2 × 3,500 = 7,000 → UGX 1,000 optional allowance remains. Plan B: marker 3,500 + extra poster 2,000 = 5,500 → UGX 2,500 remains. Both are affordable; choose based on the event's needs.

- List non-negotiable requirements.
- Calculate all-in cost, including reserve.
- Reject a plan that exceeds the budget.
- Compare the remaining feasible options using a stated goal.

Try together

Plan C adds three markers at UGX 3,500 each. Is it affordable after required supplies and reserve?

_____

_____

If the reserve increases to UGX 5,000, is Plan A still affordable?

_____

_____

Where could we use this?

A surprise price change arrives. Students recompute and defend a revised plan in writing.

A useful sentence starter

"I chose this method because _____. I checked it by _____."

Words to remember

budget · required purchase · optional purchase · reserve · constraint · trade-off

WEEK 3 / SESSION 1 / YPS-F1-U1-L05

Equal Groups and Whole Packets

What does a remainder mean when buying whole packets?

I can …

- Model equal groups using repeated addition and multiplication.
- Divide a quantity by packet size and interpret the remainder.
- Round UP to the next whole packet when partial packets cannot be purchased.

Learn the idea

Division can tell us how many groups fit. A remainder means some items are still needed. If packets cannot be split, buy enough whole packets to cover the requirement.

Worked example — follow the reasoning

18 visitors each need 1 cup. A packet contains 4 cups: 18 ÷ 4 = 4 remainder 2. Four packets provide only 16 cups, so buy FIVE packets (20 cups), leaving 2 unused.

- Draw equal groups of four cups.
- See whether all 18 visitors have a cup.
- Add one more packet for the remaining two visitors.
- Check: 5 × 4 = 20 ≥ 18.

Try together

For 23 visitors, how many 5-cup packets are needed? How many cups remain?

For 18 visitors, how many 6-cup packets are needed?

Where could we use this?

Plan supplies for a sports club with attendance changes; each learner draws groups.

A useful sentence starter

“I chose this method because _____. I checked it by _____.”

Words to remember

equal group · quotient · remainder · whole packet · total cost · feasible

WEEK 3 / SESSION 2 / YPS-F1-U1-L06

Comparing Value for Money

What is the cheapest feasible way to supply enough cups?

I can …

- Calculate the total number of packets needed for a target quantity.
- Compare total purchase costs, not just price per packet.
- Use evidence to recommend a feasible purchasing plan.

Learn the idea

Do not compare only the sticker price. Count the number of whole packets required, calculate the total cost for each option, and then compare the plans.

Worked example — follow the reasoning

We need 18 cups. A: 4 cups/packet for UGX 2,000 → 5 packets × 2,000 = UGX 10,000 (20 cups). B: 6 cups/packet for UGX 2,700 → 3 packets × 2,700 = UGX 8,100 (18 cups). Plan B costs UGX 1,900 less.

- Find whole packets needed for each option.
- Multiply packets by price per packet.
- Check each option meets the required quantity.
- Compare total costs and state a justified conclusion.

Try together

If 24 cups are needed, calculate the cost of A and B.

For 19 cups, which single-type option costs less?

Where could we use this?

Learners compare for 18 visitors and again for 19 visitors; note why answer may reverse.

A useful sentence starter

“I chose this method because _____. I checked it by _____.”

Words to remember

equal group · quotient · remainder · whole packet · total cost · feasible